

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

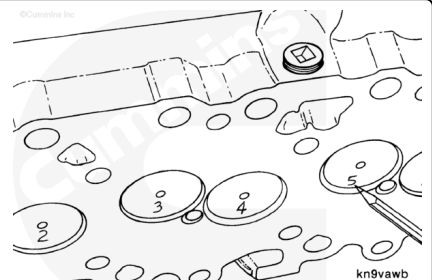
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Remove the cylinder head. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/41/41-002-004-tr.html>)



Disassemble

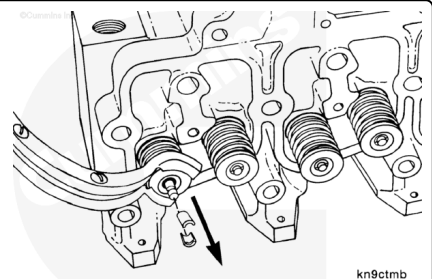
Mark the valves to identify their location.



Note : If the valve springs, collets, retainers, and valve guide seals are to be used again, they must be installed in their original location.

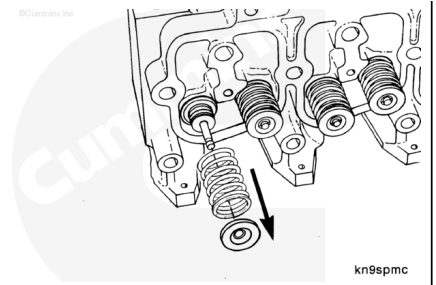
Compress the valve springs with valve spring compressor, Part Number 3375960.

Remove the valve stem collets.



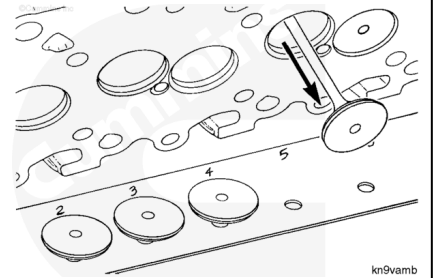
Release the valve spring and remove the valve spring retainer and valve spring.

Remove the valve.

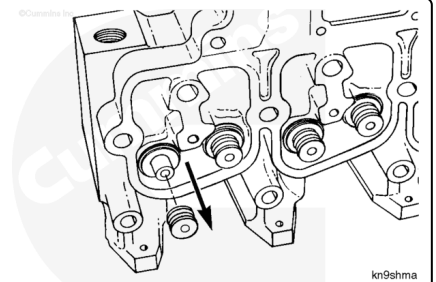


Note : Keep the valves in a labeled rack for a correct match with the companion seats while inspecting the cylinder head.

Remove the remaining collets, retainers, springs, and valves.



Remove the valve stem seals.



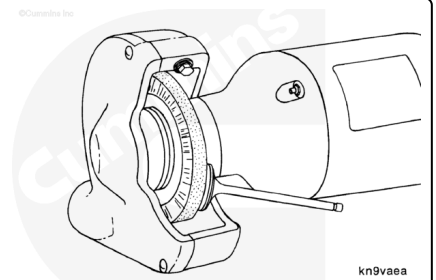
Clean

⚠ WARNING ⚠

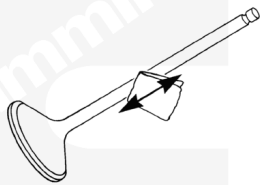
To reduce the possibility of personal injury, wear goggles and protective clothing.

Clean the valve heads with a soft wire wheel.

Note : Keep the valves in a labeled rack to prevent mixing prior to making measurements.



Polish the valve stem with a Scotch-Brite™ 7448 abrasive pad, Part Number 3823258, and solvent.



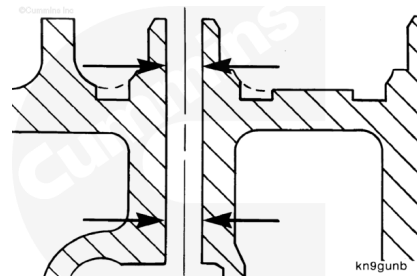
kn9vaeb

Inspect for Reuse

Inspect the valve guides for scuffing or scoring.

Measure the valve guide bore.

Valve Guide Bore Diameter		
mm		in
15.988	MIN	0.6294
16.000	MAX	0.6299

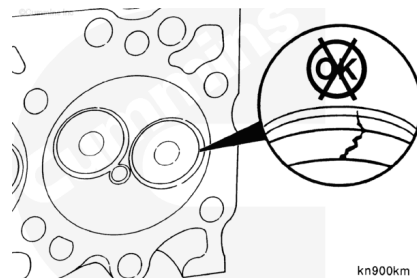


kn9gunb

If the inspection reveals damaged valve guides, reference the Alternative Repair Manual, Bulletin 3810234.

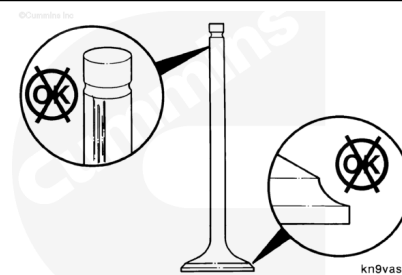
Inspect the valve seats for cracks or burned spots.

Refer to the following reuse guidelines for any cracks discovered. Service valve seats are available for seats with burned spots that will require more than 0.254 mm [0.010 in] grinding to clean up. Reference the C Series Alternative Fuel Shop Manual, Bulletin 3810234, for valve seat installation procedures.



kn900km

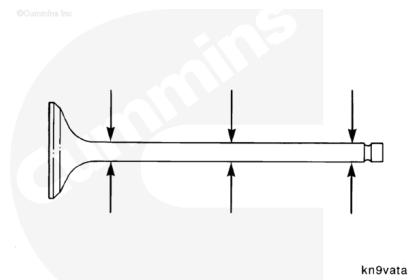
Inspect for abnormal wear on the head and stem of the valves.



kn9vasc

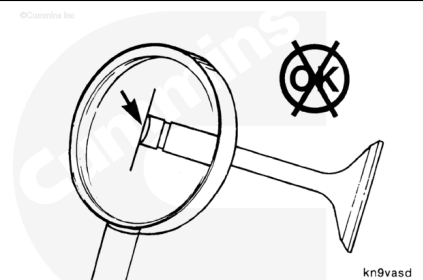
Measure the valve stem diameter.

Valve Stem Diameter		
mm		ft-lb
9.48	MIN	0.373
9.50	MAX	0.374

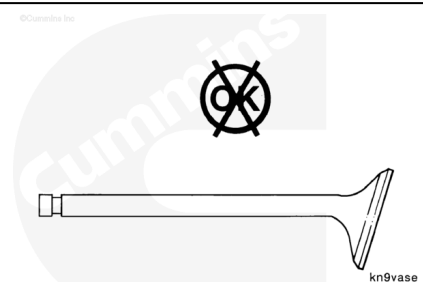


Note : If the valve is **not** within specification, it must be replaced.

Check the valve stem tip for flatness.

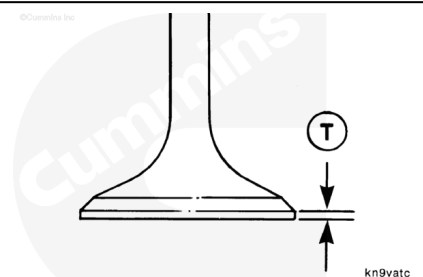


Inspect for bent valves.



Measure the rim thickness to determine if there is enough stock to grind the valve.

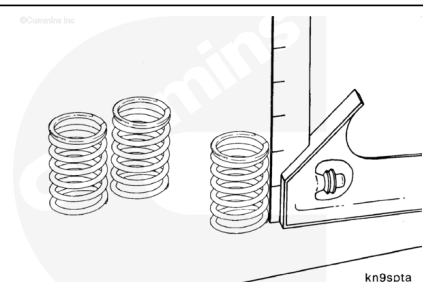
Valve Rim Thickness Limit		
mm		in
2.21	MIN	0.087



Note : If the valves are determined to be suitable for resurfacing, reference the Shop Manual, C Series Engines, Bulletin 3666008.

Inspect the valve springs.

Measure the valve spring.

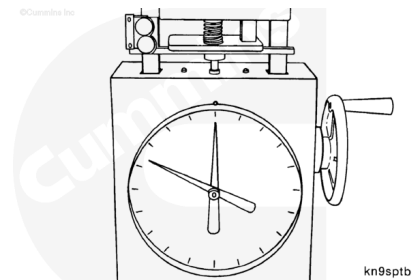


Measurements

		mm	in
Approx. Free Length (L):	1996	65.66	2.585

Measure the valve spring load.

Valve Spring Load			
	n.m		ft-lb
@ 49.25 mm [1.94 in]	359	MIN	80.7
	397	MAX	89.2

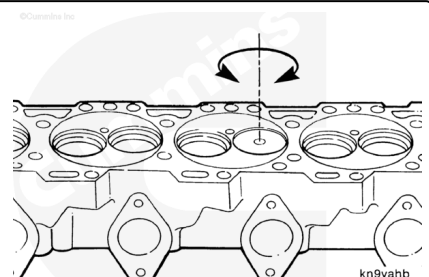


Lap

Apply a thin, even coating of fine lapping compound, Part Number 3375805, on the valve face.

Note : Use a power or hand suction lapping tool to provide pressure in the center of the valve.

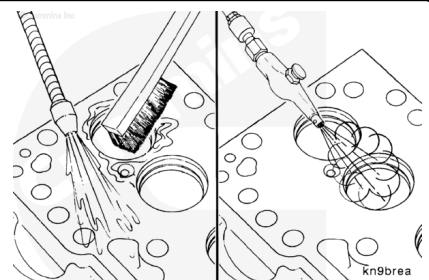
Turn the valve back and forth. Continue lapping the valve until the compound shows a continuous contact pattern on both the valve face and valve seat.



⚠ CAUTION ⚠

Lapping compound is an abrasive material. Failure will result if the cylinder head, valves, and valve seats are not cleaned thoroughly.

Clean the lapping compound from the cylinder head, valves, and valve seats.

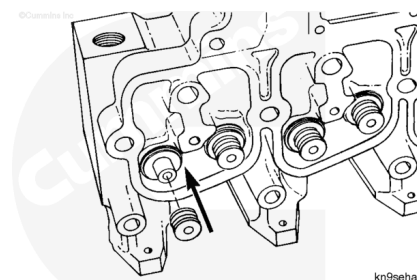


Assemble

Note : All cylinder head components must be clean before installing them.

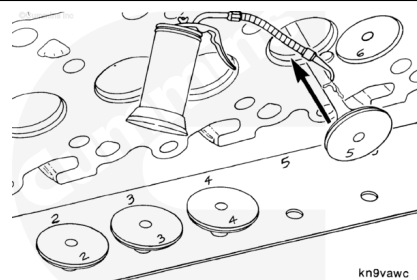
Install the valve stem seals.

Note : The intake and exhaust valve stem seals are the same.



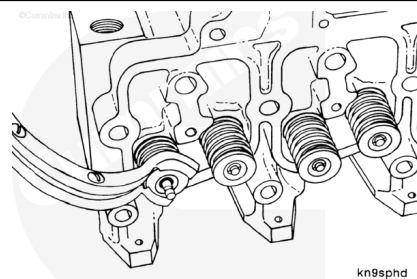
Lubricate the valve stems with clean SAE 90W oil.

Install the valves in their original locations.

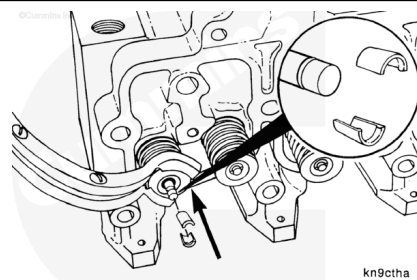


Assemble the valve spring and retainer.

Use valve spring compressor, Part Number 3375960 to compress and install the valve spring retainer assembly.



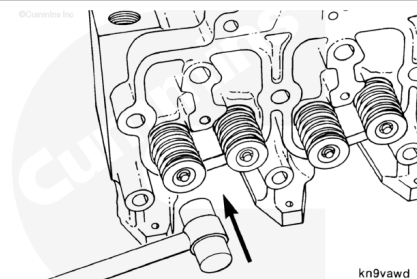
Install the valve collets and release the valve spring compressor tension.



⚠ WARNING ⚠

To avoid personal injury wear eye protection when installing valve collets. If the collets are not correctly installed, they can fly out when the stems are hit with the hammer.

After assembly, hit the valve stems with a plastic hammer to make sure that the collets are installed properly.



Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



ck800wa

- Install the cylinder head and gasket. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/41/41-002-004-tr.html>)
- Install the rocker levers. Refer to Procedure 003-008 in Section 3. (</qs3/pubsys2/xml/en/procedures/100/100-003-008.html>)
- Adjust the valve lash. Refer to Procedure 003-004 in Section 3. (</qs3/pubsys2/xml/en/procedures/41/41-003-004.html>)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/100/100-003-011.html>)
- Start and operate the engine to check for leaks and proper operation.